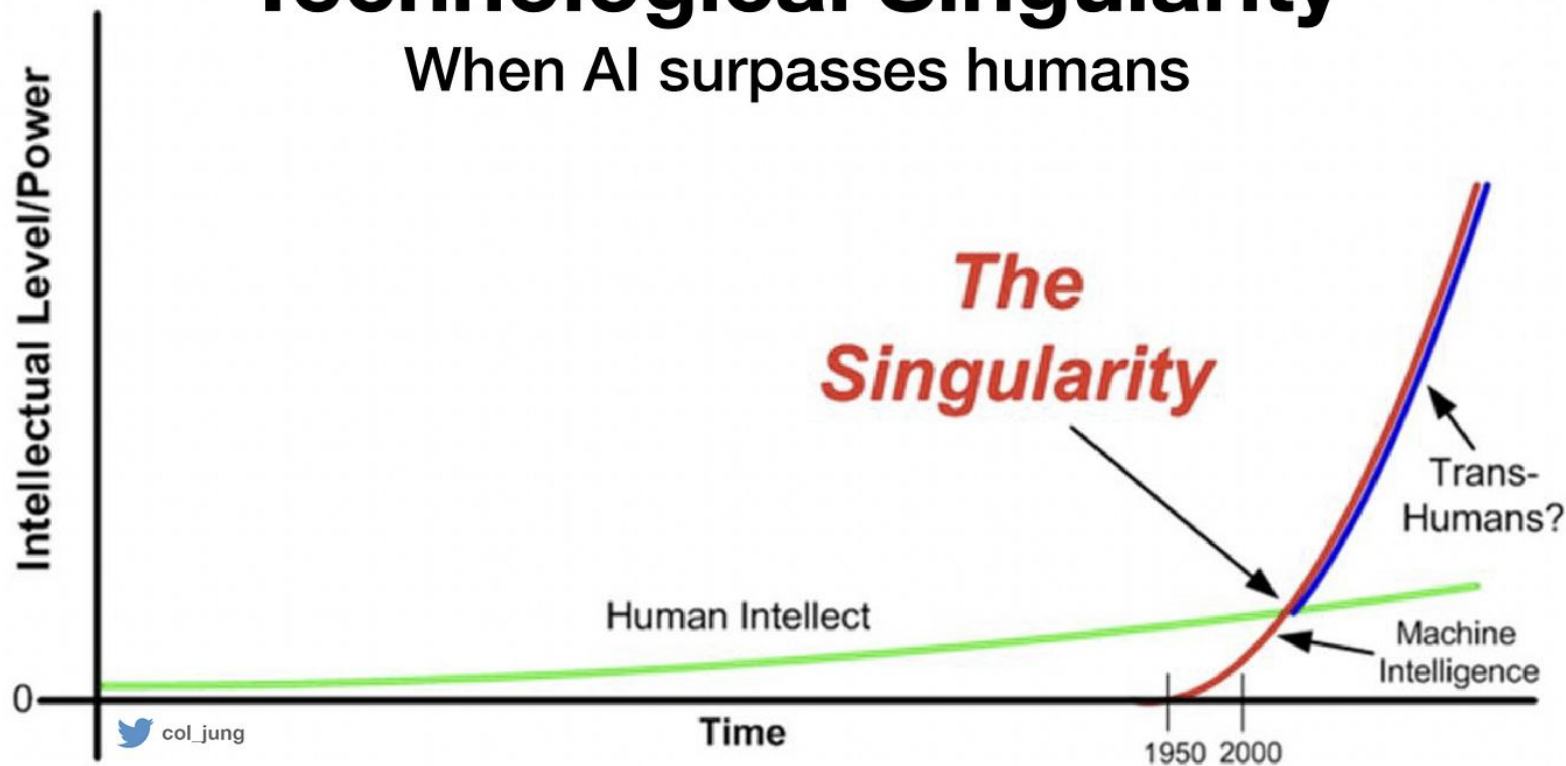




# Technological Singularity

When AI surpasses humans



# Announcements



**NEWS**

- **Project presentations**

- 20 mins to present; focus ONLY on items in the rubric
- Final slides due 5/11; should include the whole project

- **Quiz #3 on 4/22**

- Includes content taught up to today

## IS365: TASK #2 PRESENTATION SCHEDULE

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### Apr 27

#### Team 6

- 6 Nguyen,Thanh Thien
- 6 Samayoa,Liz
- 6 Thai,Annie
- 6 Kim,Mabel Grace
- 6 Victoria,Angel

#### Team 9

- 9 Lock,Nathan Russell
- 9 Fowler,Robert Jones
- 9 Echols,Bryce A
- 9 Previlon,Chelsie Pamela

#### Team 10

- 10 Trujillo Ortiz,Briana Vanesa
- 10 Duarte,Samuel Alexander
- 10 Hanson,Lauren Rebecca
- 10 Garza,Nico Luis

### Apr 29

#### Team 7

- 7 Tinajero,Tony
- 7 Li,Dylan Yen
- 7 Bayewumi,Tito David
- 7 Tang,Ryan

#### Team 2

- 2 Bacon,Angelina
- 2 Llamas-Cruz,Alex
- 2 Radford,Sam Jade
- 2 Tayam,Jelly Ann Manliguez

#### Team 5

- 5 Henriquez,Benji Orlando
- 5 Encinas,Kade Alexander
- 5 Carrera-Munoz,Dalia Belen
- 5 Ingle Awong,Kate Leinaala

### May 4

#### Team 3

- 3 Rogers,Ryan Erald
- 3 Sferrazza,Giovanni
- 3 Cabanero,Thomas D
- 3 Jones,Ryan

#### Team 11

- 11 Claudio,Kirra
- 11 Tordoff,Gregory
- 11 Reynolds,Jackson Romstad
- 11 Ayala,Yadira
- 11 Banga,Meha

#### Team 4

- 4 Boulom,David David
- 4 Chew,Mason
- 4 Otey,Peyton Aric

### May 6

#### Team 8

- 8 Turngren,Thomas
- 8 Bassi,Anthony Celestino
- 8 Wu,Brian
- 8 Weber,Zachary Lee

#### Team 1

- 1 Zuniga,Lance Nino
- 1 Mazeedian,Arman John
- 1 Lambert,Jasmine Rae
- 1 Navarro,Brogan Lucas

# Project presentation rubric

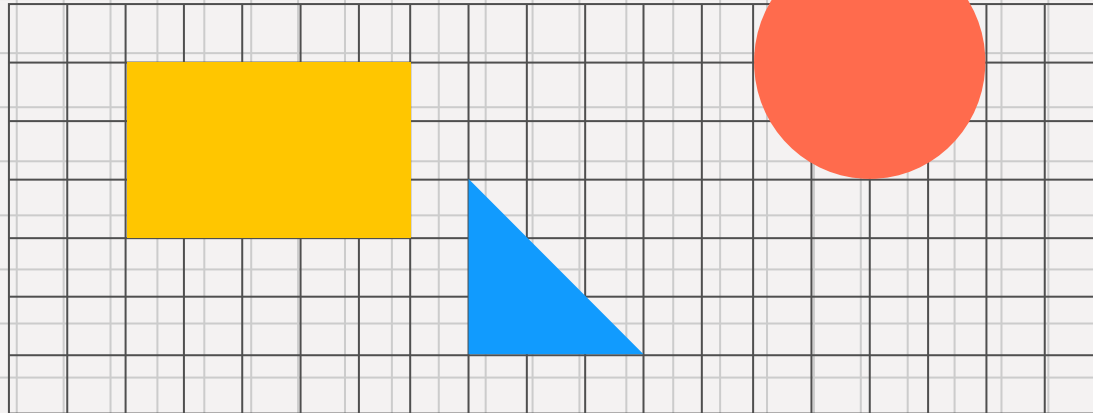
25 points possible

| CRITERION                   | DESCRIPTION                                                                                                           | SCORE                | MAX |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------|-----|
| CONTENT & SOLUTION (15 PTS) |                                                                                                                       |                      |     |
| Scope                       | Uses the "TO-BE" model to explain which parts of the process are being automated.                                     | <input type="text"/> | 2.5 |
| Tool choice                 | Outlines tools evaluated and provides clear rationale for the final selection.                                        | <input type="text"/> | 2.5 |
| Demo                        | Detailed live demo clearly showing inputs, AI tools used, and outputs.                                                | <input type="text"/> | 2.5 |
| ROI                         | Provides evidence of business impact via testimonials or return-on-investment analysis.                               | <input type="text"/> | 2.5 |
| Challenges                  | Explains 2–3 major challenges encountered and how they would approach them differently.                               | <input type="text"/> | 2.5 |
| Ethics                      | Explains specifically how humans augment, validate, and/or control the automation.                                    | <input type="text"/> | 2.5 |
| PRESENTATION (10 PTS)       |                                                                                                                       |                      |     |
| Presentation quality        | Each member states their section clearly, ends with a summary, speaks with confidence, and does not read from slides. | <input type="text"/> | 5.0 |
| Peer feedback               | Score based on survey feedback received from peers in the class.                                                      | <input type="text"/> | 5.0 |
| Total                       |                                                                                                                       | —                    | 25  |

Spring 2026

# IS 365

## **Lecture 12:** Business Process Automation Demo

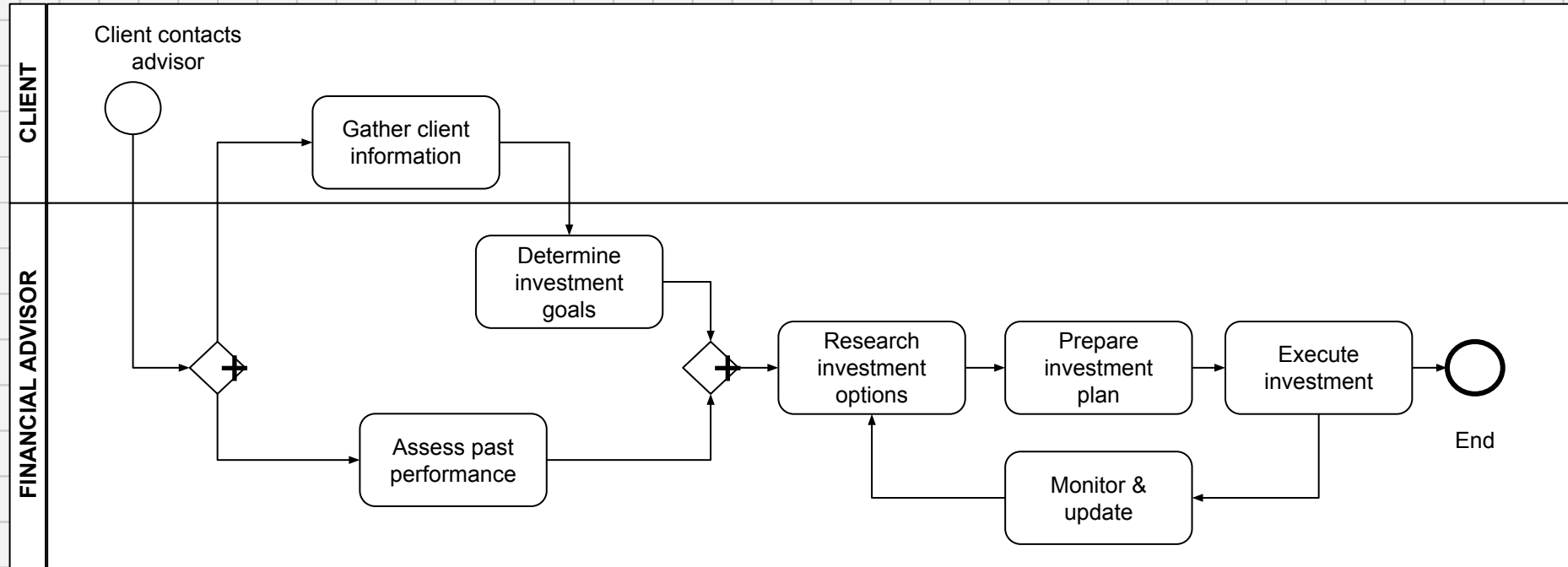


## **1. Process Description**

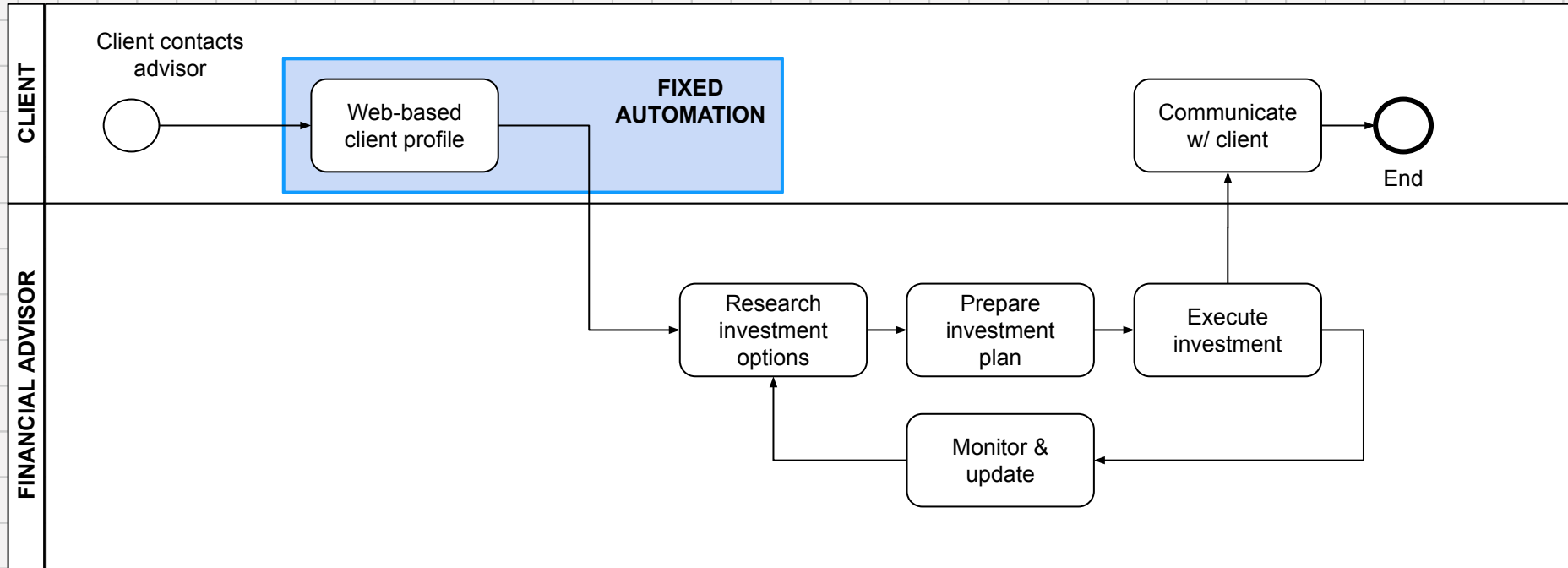
The investment advisory process begins when a client contacts the advisor. The advisor gathers and verifies financial and personal information to understand the client's profile and goals.

The advisor then researches investment options and prepares a personalized investment plan. Finally, they execute the investments and continue to monitor and report on portfolio performance regularly.

## 2. BPMN Diagram (AS-IS)

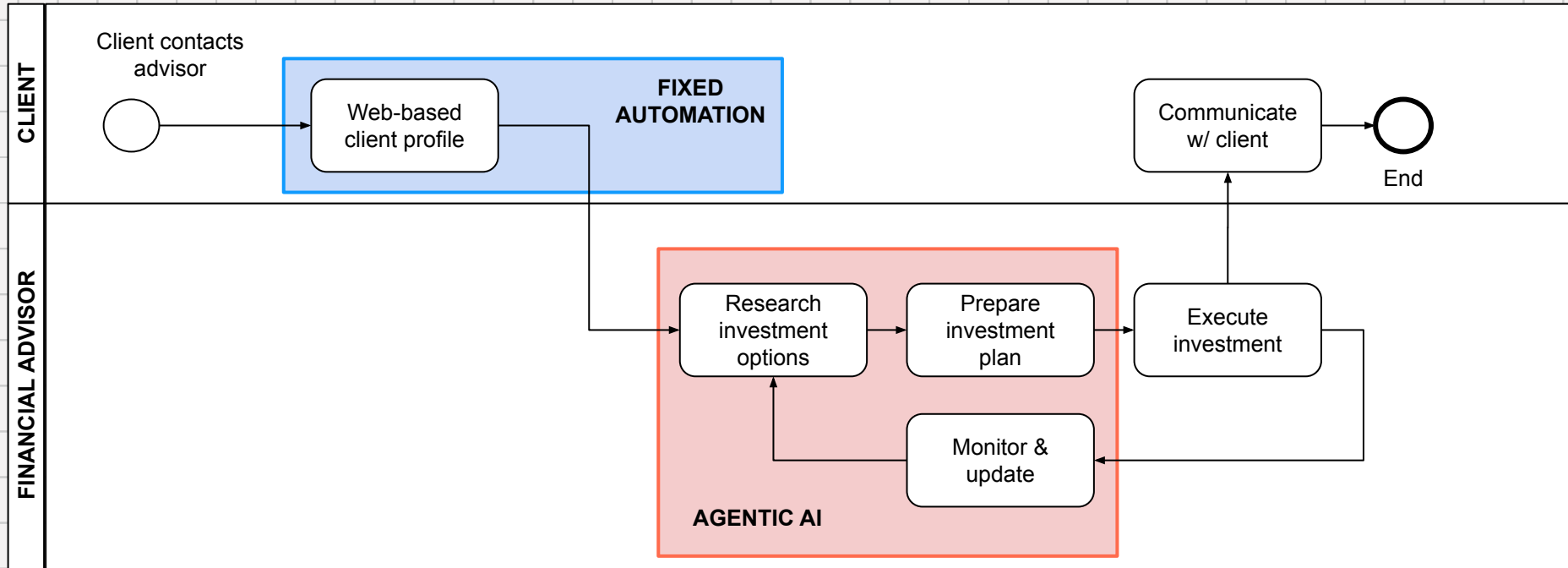


### 3. BPMN Diagram (TO-BE)

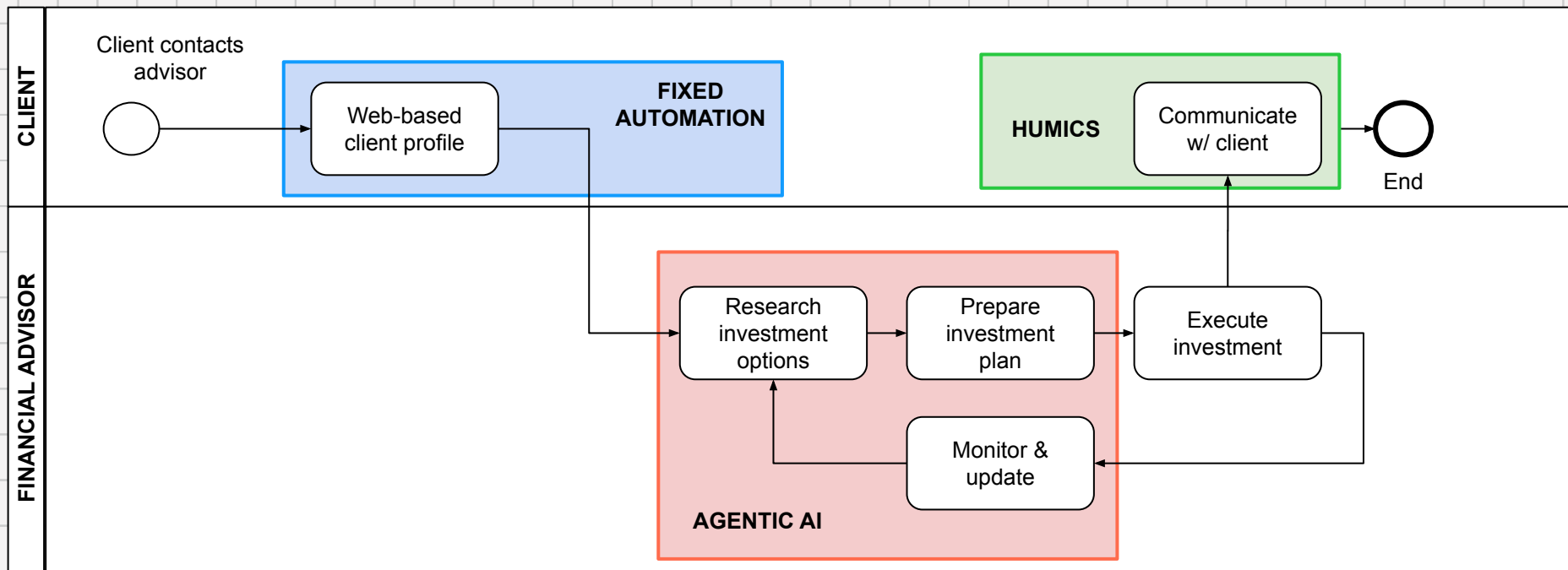




### 3. BPMN Diagram (TO-BE)

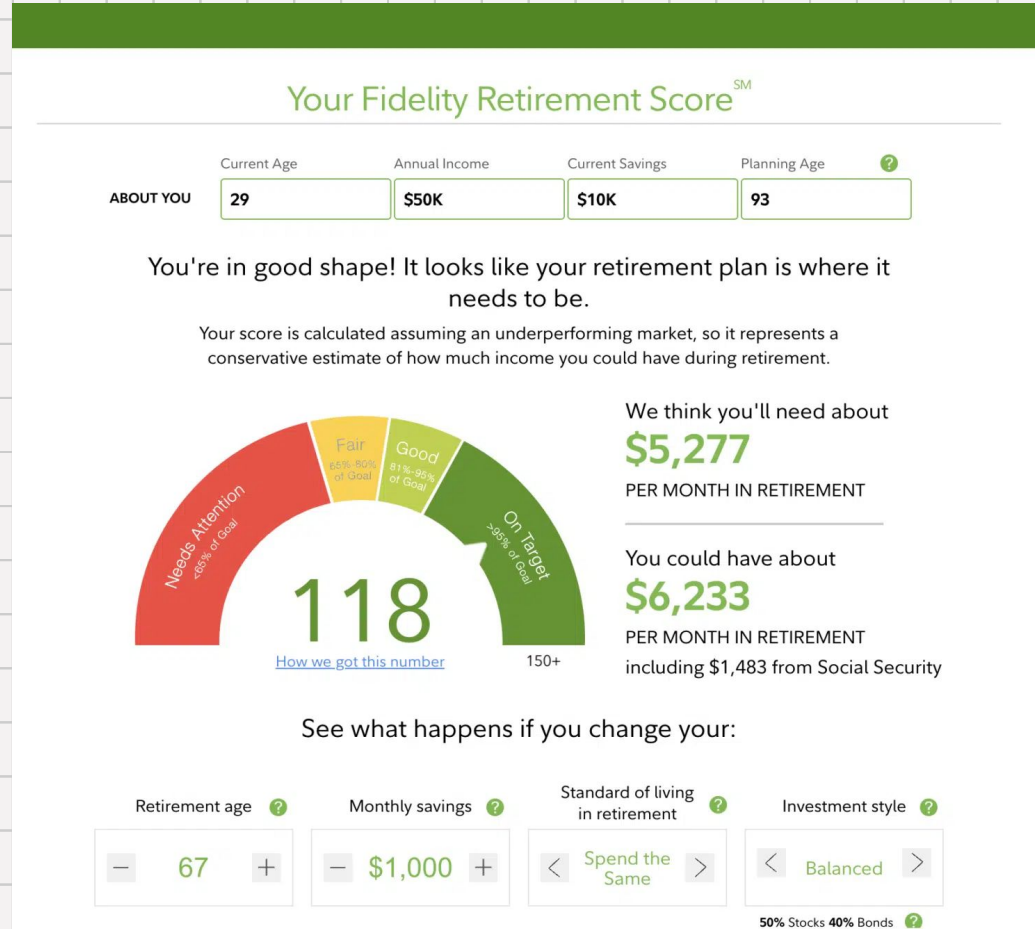


### 3. BPMN Diagram (TO-BE)



## 4. Fixed Automation

- Gamified environment
- Gathers client info, investment goals, and past performance
- Leverages data from other similar clients to determine goals



## 5. Human-in-the-Loop

- Strategy validation
- Investment execution
- Client communication



## 6. AI Agent: Research Options

### PROMPT:

Identify 5 stocks listed on NASDAQ that have shown strong performance over the last decade. Provide a short description of each stock (ticker, business focus, and why performance has been strong, historical CAGR, forecasted CAGR in 5 years) as a summary table.

| Ticker                  | Business focus                                                                                  | Why performance has been strong (past decade)                                                                                                                                                                                                                    | 10-yr historical CAGR<br>(price / total return)                                                  | Forecasted CAGR (next 5 yrs)                                                                                                                  |
|-------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| AAPL — Apple Inc.       | iPhones, Mac, Services (App Store, iCloud, services/subscriptions).                             | Enormous hardware scale + higher-margin Services & ecosystem lock-in drove steady margin expansion and strong multiples. <a href="#">Techloy +1</a>                                                                                                              | ~24.5% (10-yr price CAGR). <a href="#">FinanceCharts</a>                                         | ~9–9.5% p.a. (5-yr EPS / revenue growth forecasts from model/consensus). <a href="#">Finbox +1</a>                                            |
| MSFT — Microsoft Corp.  | Enterprise software, Azure cloud, Office/365, AI products (Copilot), LinkedIn, server products. | Rapid cloud (Azure) adoption + AI monetization & recurring subscription revenue have driven revenue & earnings compounding. <a href="#">Source +1</a>                                                                                                            | ~27% (10-yr total/price CAGR). <a href="#">FinanceCharts</a>                                     | ~12–15% p.a. (5-yr revenue / EPS growth consensus / forecast). <a href="#">Finbox +1</a>                                                      |
| NVDA — NVIDIA Corp.     | GPUs and AI/data-center accelerators, software ecosystem (CUDA, AI stack).                      | Dominant position in AI training/inference GPUs and data-center accelerator market created explosive revenue and earnings growth. <a href="#">The Chronicle-Jo... +1</a>                                                                                         | ~75% (10-yr price CAGR — extremely high, reflecting AI-era surge). <a href="#">FinanceCharts</a> | ~20–30%+ p.a. (5-yr revenue / EPS forecasts vary by model; many models project strong double-digit growth). <a href="#">Finbox +1</a>         |
| AMZN — Amazon.com, Inc. | E-commerce + Amazon Web Services (AWS) cloud + advertising.                                     | Scale in e-commerce + rapid growth and high margins from AWS and advertising steadily boosted profitability and valuation. Recent AI/cloud tailwinds amplified AWS growth. <a href="#">Reuters +1</a>                                                            | ~23% (10-yr price CAGR). <a href="#">FinanceCharts</a>                                           | ~18–22% p.a. (5-yr EPS / revenue growth forecasts in many models). <a href="#">Finbox +1</a>                                                  |
| TSLA — Tesla, Inc.      | Electric vehicles, energy storage & solar, Autonomy/Software.                                   | Extremely rapid vehicle production growth, margin improvement in key years, plus investor enthusiasm for EV leadership and  gy/storage upside. <a href="#">Investing.com +1</a> | ~39–41% (10-yr price / total-return CAGR — very large). <a href="#">FinanceCharts +1</a>         | ~30–40% p.a. (Finbox/analyst EPS & metric forecasts are aggressive; models show high multi-year EPS/revenue CAGRs). <a href="#">Finbox +1</a> |

## 6. AI Agent: Simulate Plan

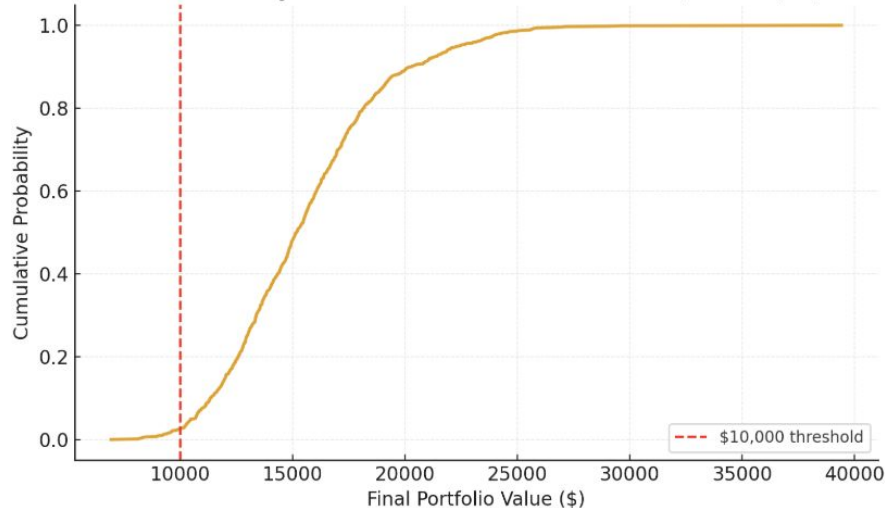
### PROMPT:

You are a Monte Carlo simulation engine. I invest \$5,000 equally across the 5 stocks you've identified. Assume the forecasted CAGR and annual volatility of 20%. Simulate 1,000 random 5-year future paths for the total portfolio value. Show the output as a probability chart.

Provide:

1. The probability the portfolio value reaches or exceeds \$10,000 in five years.
2. The median outcome and 90th percentile outcome.
3. A plain-language summary of what this means.

Cumulative Probability Distribution of Portfolio Value (5 Years, 1,000 Sims)



Here's the **cumulative probability chart** for the 5-year Monte Carlo simulation — it shows how likely your portfolio is to reach or exceed different ending values.

- The **red dashed line** marks the \$10,000 threshold.
- The curve shows, for example, that around **97%** of simulated outcomes ended **at or above \$10,000**.
- The steeper section around \$14–16k marks the most likely range of results, while the flatter tail to the right represents rare high-growth outcomes (top 10–20% scenarios). </>



## 6. AI Agent: Monitor & Report

**PROMPT:**

I invested \$5,000 equally across the 5 stocks you've identified on Jan 1, 2024. Provide a summary table with the first column representing the starting value, and then additional columns representing value after every 6 months. The rows should show each individual stock and the last row the total portfolio value.

| Stock | 0mo     | 6mo     | 12mo    | 18mo    | 24mo    |
|-------|---------|---------|---------|---------|---------|
| AAPL  | 1000.00 | 1108.21 | 1122.65 | 1270.98 | 1628.53 |
| MSFT  | 1000.00 | 974.79  | 949.90  | 1023.08 | 812.40  |
| GOOGL | 1000.00 | 1312.95 | 1357.09 | 1462.13 | 1275.58 |
| AMZN  | 1000.00 | 960.70  | 1305.85 | 1363.36 | 1227.97 |
| TSLA  | 1000.00 | 1214.62 | 1361.61 | 1465.67 | 1536.85 |
| Total | 5000.00 | 5571.27 | 6097.11 | 6585.22 | 6481.34 |

## 6A. Agentic AI w/ Orchestration

- Multiple agents working autonomously

**PlannerAI:** Let's aim to grow \$1,000 to \$10,000 in 5 years. I suggest we diversify across 5 NASDAQ stocks, balancing growth potential and risk. I can propose an initial shortlist.

**ExecutorAI:** Sounds good. Please share your shortlist. I'll analyze historical performance, volatility, and 5-year CAGR to see which ones are most likely to achieve our goal.

**PlannerAI:** Based on past performance and market trends, I propose:

1. Tesla (TSLA)
2. Nvidia (NVDA)
3. Apple (AAPL)
4. Amazon (AMZN)
5. Microsoft (MSFT)

**ExecutorAI:** I'll run a simple Monte Carlo simulation assuming average growth rates and historical volatility to estimate 5-year outcomes: Simulation results:

- 70% chance of exceeding \$5,000
- 50% chance of reaching \$10,000
- Some risk of falling below \$3,500

**Agreed.** I suggest we allocate: TSLA 25%  
NVDA 25% - AAPL 20% - AMZN 15% - MSFT 10%

**ExecutorAI:** Agreed. Final investment plan:

- TSLA 30% - NVDA 30% - AAPL 15%
- AMZN 15% - MSFT 10%

Review biannually and rebalance as needed